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RESEARCH ARTICLE–ELECTRICAL ENGINEERING

Mel-Frequency-based Feature Analysis of Audio Signals in the Context of Holy Quran Recitation

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Abstract

Different sounds have various effects on human health, and by introducing the ones that are therapeutic, a healing environment can be created. This paper describes the process to train and test a machine learning algorithm to describe and explore the therapeutic nature of Quranic verse. Using a dataset containing four emotional states namely happy, sad, angry, and relaxed, we trained a model and classified different recitations of the Quran into one of these states. This paper proposes the use of Mel-frequency cepstral coefficients (MFCC) to extract features from Quranic audio and classify it with respect to a known dataset. Based on the experiments conducted on Quranic verses, we summarize our results.

Keywords MFCC · Cepstrum · Deep-learning · Mel-spectrum · Therapeutic · Holy Quran recitation · Audio analysis

1 Introduction

This paper investigates as to why listening to Quranic verses is considered healing and/or therapeutic. "Be aware that the remembrance of Allah calms the hearts" is one of the verses in the Quran about achieving tranquility. Quranic recitation therapy has been reported to have cured patients having different ailments and therefore has globally sparked an interest in figuring out whether these verses display therapeutic properties [1]. For a segment of audio to be therapeutic, it should be classified as soothing for a patient, as soothing sound is regarded as good for human health. There are various ways with which this classification can be made and one of those ways is to conduct a survey or to utilize sensors to examine any psychological or physical changes [2]. This approach is tedious and is prone to participants being biased. Belonging to a certain demographic, ethnicity, and religion, the observers tend to be biased and can make decisions based on nostalgia and/or upbringing. Similar studies have been conducted before in which the data was collected from different individuals, but the participants were Muslims, and therefore that data cannot be classified as unbiased [3]. In this paper, we utilize a dataset that was composed of audio samples based on four different emotional states, i.e., relaxing, happy, sad, and angry [4]. We performed deep learning processing on multiple Quranic verses in order to analyze their effect on human health.

1.1 Effects of Quranic Verses on Mental Health

Anxiety is the most common among all psychological disorders that has both pharmacological and non-pharmacological solutions. It can be defined as a feeling of apprehension, stress on nerves, nervousness and worry accompanied by psychological discomfort. Pharmacological interventions like selective serotonin re-uptake inhibitors (SSRIs), while showing greater effectiveness initially, do not present a lasting remedy. Using drugs to manage anxiety can lead to low productivity and other numerous side effects. Non-pharmacological solutions like aromatherapy, massage therapy, relaxation techniques and music therapy are much more stable and have minimal side effects [5]. In numerous studies, music therapy has been proven to have anxiolytic effects on patients suffering from cancer and Alzheimer's. Religious audio is

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garnering recognition worldwide as it has proven to have stress-relieving effects. Anxiety management in various settings using religious therapy is a growing interest among researchers. Relevant studies from databases like SID, Iranmedex, Mairan, PubMed, Scopus, and Google Scholar have underlined the significance of the effect that is caused by these stimuli. Recitation of the Holy Quran has been attributed to the release of endorphin hormones by stimulating alpha brain waves. The data in various studies indicate that Quranic verses do in fact help with anxiety [6, 7]. In one study, the effect on mental mean health between the test group (those who listened to the recitation) and the control group (those who did not listen to Quranic verses) was evaluated where it was found that the difference between mean mental health (P) of the two groups is around 0.037, i.e., a similarity of less than 4% [8].

Getting admitted to a hospital is stressful where 85% of people in the intensive care unit (ICU) are believed to have symptoms of distress. Therefore, the type of sound that they are fed can play an important role in reducing stress levels [9]. In a study, two almost identical groups were selected. One of these groups was given the conventional treatment and the other one was given a 30 min session of Holy Quran recitation (HQR). This process was evaluated by subjecting the participants to weaning. The physiological parameters like rapid shallow breathing index, respiratory rate, heart rate, oxygen saturation, exhaled carbon dioxide, and blood pressure were recorded. The study yielded that HQR had no negative impacts and did not hinder the process of weaning. This study proved itself to be a testament to the fact that Quranic recitation can be used as therapy for recovering patients [10].

1.2 Related Research

There are multiple factors that can categorize audio to be soothing or relaxing. Some audio segments can be classified as soothing, because of different vibrations that are produced and in some cases, emotions can be conveyed via music by composing music that mimics the functionality of pulses of a brain [11]. Sound can be classified into different categories based on different features. The ones that are most prominent are pitch, frequency, time period, trend, timbre, loudness, and echo. Although a valid assumption can be made about the given audio, we must account for the fact that when the analysis becomes too specific, the corresponding number of classes increases. As a result, differentiating different classes based on just these characteristics becomes hard, and the assumptions become inaccurate. Therefore, mere assumptions cannot be relied upon. To classify certain audio into a given class where the audio directly impacts human chemistry, we must create a robust system on which we can rely to make accurate predictions.

One of the ways in which audio can be classified into a certain category is by implementing multiple sensors to observe changes in brain activity, skin conductance, and heart rate. By doing so, certain characteristics of the audio and its effect on human subjects can be deduced. By examining the response of the subject to given stimuli, the audio can be classified into different categories and can be differentiated from other classes. The fundamental problem with this approach is that this heavily relies on the test subjects and the data gathered can be biased. In the scenario where participants have a religious background, certain religious audios will have some impact on them due to nostalgia and religious beliefs [3]. In the studies with unbiased data, the approach that was used to classify audio was to gather all the information extracted by the participants and create a dataset [12]. Afterward, machine learning can be utilized to classify any given audio in a category from the set of categories [13].

Different machine learning approaches having different types of data can be utilized to make the classification. The approach will largely depend on the type of data and features used. The analysis can be done on audio directly, or it can be done by transforming into another domain. For image processing, different characteristics can be plotted, and then the model can learn the difference between all the classes. In this approach, the difference between all the classes can be found with the help of the convolutional neural network (CNN) [14]. In the other case, analysis can be done directly on audio, which will yield a lot of data regarding the audio, which can be done using Mel-frequency cepstral coefficients (MFCC) [15]. We can extract similar features from the chroma-spectrogram [4]. For the extraction of features, it is not a compulsion to use only one approach. Multiple approaches have been used before to make the model less prone to errors [16, 17]. With the help of extracted features, different classifications of audio, music, speech and environmental sound can be done [18, 19].

Implementing feature extraction to the recitation of the Holy Quran has been widely done before. Feature extraction has proven itself to be a very practical tool as there are a lot of different recitation techniques which can be processed with ease using the extracted features. The acoustic features in the Quranic recitation are contained uniquely in frequency. Interferences tend to exponentially decrease the accuracy of the models that are made with the help of extracted features and to prevent that, speech is normalized and warped in time series. Recent studies have shown that the most suitable technique for extracting features is MFCC. Within the MFCC techniques, the use of Warping DFT has shown improvement over conventional MFCC [20].

Our contribution is as follows:

1. Giving an insight in the MFCC.

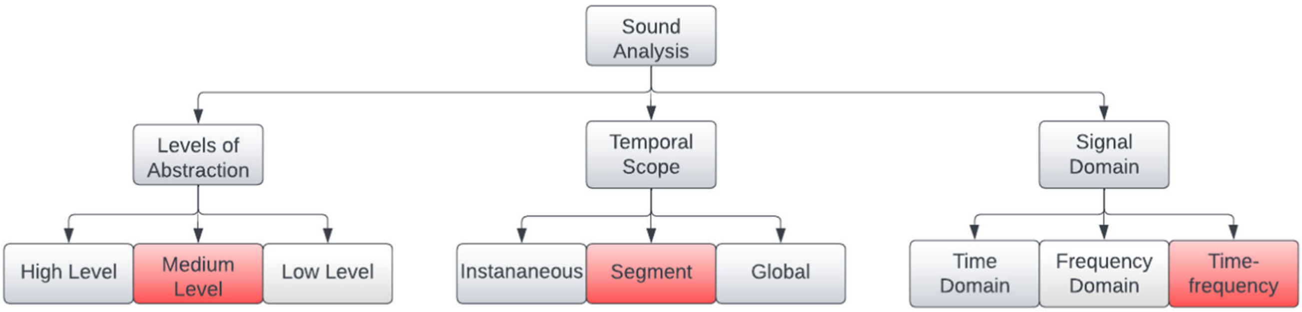

Fig. 1 Categorization of features based on different styles

2. Modeling a neural network to determine the emotion associated with an audio
3. Categorizing Quranic verses as soothing using the machine learning model

The remaining sections are organized as follows: Section 2 outlines the entire process with which the machine learning model was formulated. This section provides an insight into the different feature extraction techniques that were used along with Python code that was utilized to carry out the deep learning classification. Section 3 provides information regarding the performance of the algorithm along with the findings regarding the soothing nature of Quranic verses. Finally, Section 4 concludes the paper.

2 Methodology

In this section, the process of feature extraction and corresponding classification is discussed. Figure 1 shows the illustration of different techniques that are used for feature extraction for sound analysis.

2.1 Various Methods for Audio Classification

Various methods can be used to derive the audio features for the analysis of sound. We can classify them into three groups for sound classification. The first method is to segregate the sample into high, medium, and low-level features called levels of abstraction. The high-level features are the ones that can be perceived and understood by human beings, like the bar of a sound or its pitch. Medium levels are the features that can be perceived by humans but cannot be understood in the absence of a machine like MFCC of a sound signal. Lastly, low-level features are the ones that can only be perceived by machines like RMS energy and zero crossing rate.

The second method for sound classification is called temporal scope where the features are classified as instantaneous, segment-level, and global level. The instantaneous features give information on short chunks of audio signals as small as 10–20 ms whereas segment-level divides the audio chunks into a 1–20 s range. In global-level features, the information consists of large files like a musical phrase or a bar.

The third and most used method is the signal domain, where the features are categorized into the time, frequency, and time-frequency domain. Time domain features are extracted from a waveform in the time domain like root-mean-square energy and zero crossing rates. Just like the time domain frequency domain features are extracted from the frequency domain graph, like the spectral centroid, spectral spread, and spectral skewness. Time-frequency representation includes spectrograms and the Mel-spectrograms. This is the type of signal classification that was used in this research. Figure 1 shows the mapping of the technique used with respect to the different methods.

Some of the basic features extracted for sound analysis are root-mean-square energy (RMSE) and zero crossing rate, along with MFCC. Root-mean-square (RMS) is a mathematical method used for observing continuous power for any signal. RMS is used in audio signals for amplitude because it is constantly varying and the more complex a sound is, the harder it is to get an accurate average reading. To counter this, the values are squared, and the mean value is calculated over a period in which the square root is taken. This gives us the RMS rating as given in (1).

$$\mathrm{RMS} = \sqrt{\frac{1}{K} \cdot \sum_i {s_i}^2} \tag{1}$$

where (K) is the number of samples, and (s) is the amplitude/power of an audio signal.

RMS indicates the loudness of the sound signal in question. This is usually taken because it is less sensitive to outliers, like spikes in amplitudes of the signal. We can use RMS energy to identify new segments in an audio signal.

The zero crossing rate represents the number of times a given signal crosses the horizontal axis. We get this by comparing the amplitude values of consecutive samples. This is used for recognizing percussive or pitched sounds, as percussive sounds have unstable zero crossing rates while pitched sounds have more stable zero crossing rates.

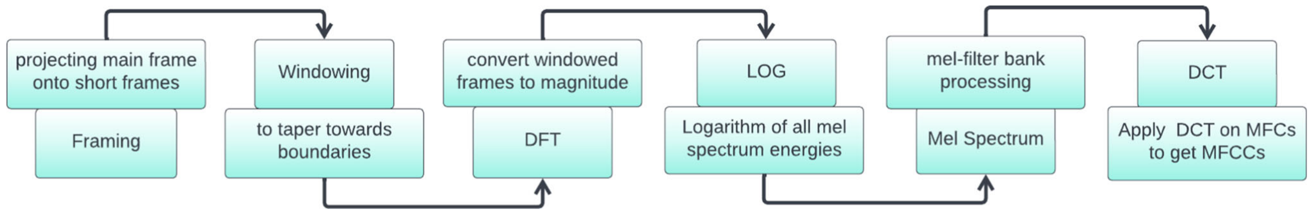

Fig. 2 MFCC stages

2.2 Mel-Frequency Cepstral Coefficients

We rely on certain features for audio classification. Some of the features are extracted by converting the audio signal into a different domain, which helps to distinguish the different components of the audio. Most of the features hence are named based on the domain in which they are generated. To make them easily understandable, words like cepstrum, quefrency, liftering, and rhamonic are used, which are derived from spectrum, frequency, filtering, and harmonic, respectively.

In 1960, cepstrum was introduced to study echoes in seismic signals, but later it was used for audio feature extraction for speech processing. Considering its wide application later, it was used in MFCC extraction. MFCC are used to extract valuable information from a given audio signal. To obtain a cepstrum, first, we must convert the time domain signal into a spectrum which is done by taking a discrete Fourier transform of the signal and getting it into the frequency domain. The next step is to convert the spectrum into a log spectrum. We take the inverse Fourier transform and instead of going into the time domain; the signal goes into the quefrency domain.

The reason for doing all this computation is rather simple. To find all the frequency components of a signal given in the time domain, we convert the signal into the frequency domain, which gives all the sinusoids that were involved in the making of that time domain signal. Similarly, we take the inverse Fourier to take the spectrum of an audio signal that is already in the frequency domain. Cepstrum can also be termed as the spectrum of a spectrum and to filter out the components that we do not need, we take the log of the spectrum. To understand this, we must give a little background. Humans produce speech through the vocal tract and the air is pushed out of the vocal cords, which results in different sounds. The change in vocal cords is the thing that is vital for human speech, and it is responsible for the production of different sounds. In a way, the amalgamation of all the components in the vocal tract act as a filter for speech. With the vocal tract, the glottal pulse gets translated into intelligible speech. The log functions are used to separate the spectral envelope and the spectral detail like the function that happens in the vocal tract. Different formants can be extracted from the spectral envelope, giving us valuable features of audio data.

By using the log, we can add the parts of the signal by using logarithmic properties instead of them being convolved. This way filtering out spectral details gets easier, and we do not have to use other techniques to separate different parts of the audio, those being the glottal part and other useful characteristics. The frequency response of the signal is the envelope and contains useful information.

The envelope of any audio signal is on the lower quefrencies and the glottal pulse is on the higher quefrencies in the quefrency domain, so we must filter out the glottal pulse, as this part does not contain any useful information. By taking the log, we already made the spatial difference between both pulses, and we can filter out the important features. The process of filtering in the quefrency domain is called liftering. By judging the different harmonics which in the quefrency domain are called rhamonic, we can analyze the exact quefrency at which the envelope lies.

The signal is divided into different parts and from all those parts multiple means are taken. Initial 12 to 13 coefficients are picked out as they contain the most relevant information. Taking more coefficients is not going to affect the accuracy of the model, as the initial coefficients contain the most relevant information. However, to increase the accuracy of the model, we can take the derivative. A signal with a little duration will have a lot of frames, and each frame contains multiple coefficients. By taking the differentiation, we subtract the next frame from the previous frame, turning the number of coefficients into either 26 or 39 coefficients by taking the first and second derivatives, respectively. MFCC work well with speech and music processing and ignore fine spectral details. Figure 2 provides all the steps that are involved in the extraction of MFCC.

More detail about individual steps in the computation of MFCC is given below.

2.2.1 Framing

In the process of framing, the different audio signals that are obtained by analog-to-digital conversion are divided into 20 to 40-ms chunks. This process is done so that each frame is composed of a single frequency. The goal is to extract segments that are short, so that the properties of the signal do not have time changes within the segment.

2.2.2 Windowing

Splitting the signal into different frames results in numerous undesirable discontinuities at the border of each frame. Windowing is the process of eliminating these discontinuities by being implemented at the beginning and the end of each frame. Hamming windowing is the most well-known type of windowing when it comes to audio processing. Hamming windowing shrinks the value of starting and ending of each frame to zero and as a result, there are no discontinuities. The formula for windowing is given in (2)

$$W(n) = W_{\circ}\left(n - \frac{N-1}{2}\right) \tag{2}$$

where $W(n)$ is the hamming window and N is the number of samples in each frame.

2.2.3 Discrete Fourier Transform

In the time domain, glottal pulse and vocal tract impulse response are combined by convolution as shown in (3).

$$x(t) = e(t) * h(t), \tag{3}$$

where $x(t)$ is the speech signal and $e(t)$ and $h(t)$ are the glottal pulse and impulse response, respectively.

To make the process of extracting the impulse response easy, we convert the signal into the frequency domain. This way, the convolution between both signals converts into multiplication as shown in (4).

$$X(f) = \text{FFT}[e(t) * h(t)] = E(f) \cdot H(f) \tag{4}$$

where $X(f)$, $E(f)$, and $H(f)$ are the Fourier transform of $x(t)$, $e(t)$, and $h(t)$, respectively. The spectrum of the audio sample can be yielded by taking the Fourier domain analysis. The spectrogram of adhan (call for prayer) is given in Fig. 3

By implementing the next few steps, it is possible to separate signals.

2.2.4 Logarithmic Property

To separate the desired signal from the glottal pulse, both signals should be summed instead of being multiplied, as shown in Eqs. (5) and (6).

$$\log[X(f)] = \log[E(f) \cdot H(f)] \tag{5}$$
$$\log[X(f)] = \log[E(f)] + \log[H(f)] \tag{6}$$

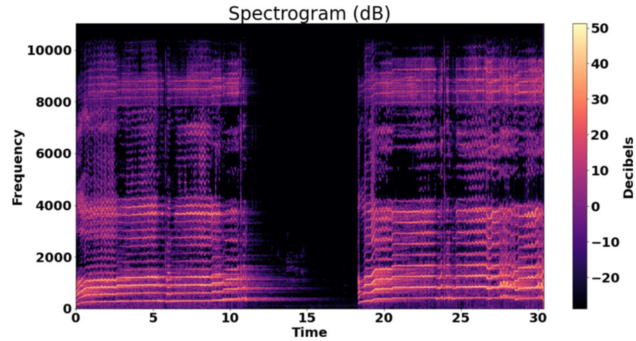

Fig. 3 Spectrogram

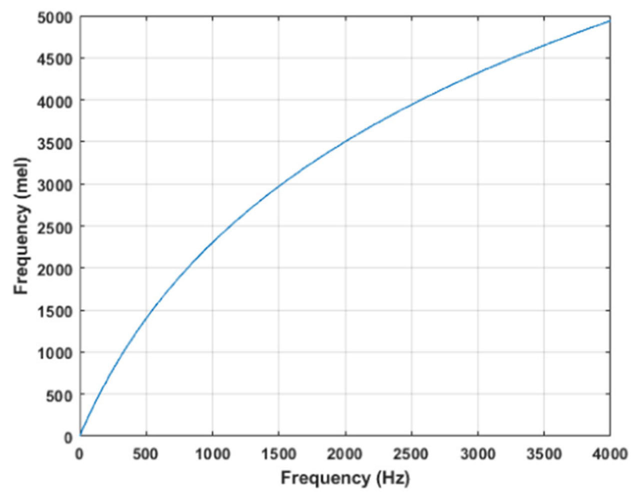

Fig. 4 Relation between Mel and Hertz

By doing so, both signals are independent of each other and can be separated by applying a filter.

2.2.5 Mel-Filter Bank Processing

The voice signals are not linear and are, instead, logarithmic. The range of the FFT becomes wide, so instead of plotting everything on the hertz scale, the signal is plotted on the Mel scale. Mel scale is the perceptually correct scale. The difference between the two frequencies is more prominent in lower frequencies with the same frequency difference as opposed to the ones on the higher end. To make the difference between both higher and lower frequencies equal, the signals on the frequency scale are plotted on Mel scale. The relation between the frequency scale and the Mel scale is defined in (7) and plotted in Fig. 4

$$\text{Mel} = 2595 * \log_{10}\left(1 + \frac{f}{700}\right) \tag{7}$$

where Mel and f are frequencies in the melody (Mel) and hertz (Hz), respectively.

Mel-spectrogram of adhan is given in Fig. 5

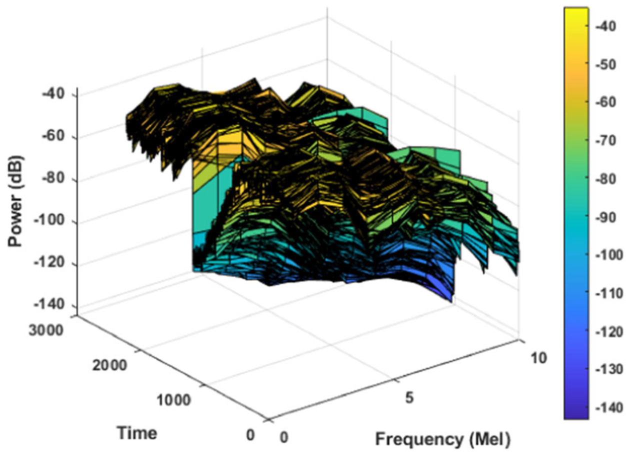

Fig. 5 Mel-spectrogram

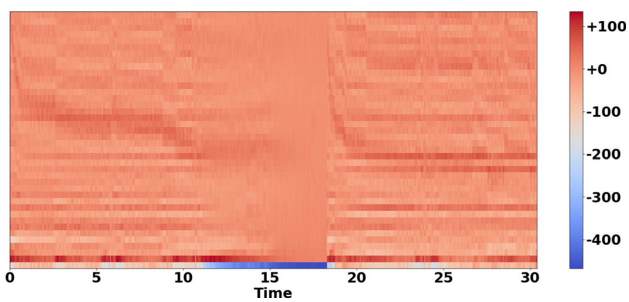

Fig. 6 MFCC graphical plotting

2.2.6 Discrete Cosine Transform

By taking discrete cosine transform information regarding different formants is extracted, resulting in multiple MFCC. By implementing this step, the difference between both signals becomes very apparent and liftering becomes possible. Graphical plotting of MFCC of adhan are given in Fig. 6.

2.3 Machine Learning Model

The dataset that was utilized contains audio that is classified into four different emotions [4]. These emotions can be experienced while listening to different music or sounds. The idea that sounds can influence and induce emotions is not new and there is a lot of data available to categorize different sounds on the emotional spectrum. The four emotions used were happiness, sadness, anger, and relaxation. Apart from metadata, the folders contained different audio, representing different emotions. There were 100 audio clips corresponding to each emotion, and each audio was 30 s long.

We used Python as it is objectively the better platform to carry out data science and artificial intelligence application. We carried the machine learning applications out through Google colaboratory. Google Colab is a much better alternative to any other python IDE. This notebook is very similar to software like Jupiter and is tailored for TensorFlow and Keras frameworks. Colab allows a user to do their machine learning development in a single notebook and the entire computation runs on the cloud on the server side. This saves us from the hassle of downloading packages and dependencies as all these tasks are done on a remote machine.

The work done on Colab makes it very easy to share the work with peers. To make accurate classifications along with making the learning process far easier, different tools were employed. The application requires large data analysis for which pandas was imported. For complex mathematical functions, NumPy was used as it possesses the ability to make multi-dimensional arrays. OS was imported as working with the dataset, multiple files are required to be added and moved to and from different directories. Matplotlib was used to display the analysis in a graphical format. Librosa was imported as the most vital library, as it is responsible for all or any operations that have anything to do with audio. Librosa is also responsible for MFCC analysis of different audios other than the ones already in the dataset.

To utilize the desired functions of MFCC, a separate function was declared named" feature extractor". This function was responsible for loading the data and extracting the features utilizing librosa. A total of 40 features were extracted. Utilizing the extracted features, the data was related to the CSV file by running as many iterations as the number of audio files. After this process, the relation was shown by arranging them into a data frame utilizing pandas. Dataset was split into dependent and independent datasets. Label encoding was done on the class dataset. Train test split was done using sklearn with an 80,20 split for training and testing, respectively.

To carry out deep learning applications, TensorFlow was used. In TensorFlow, sequential was used and in it, dense, dropout, activation, and flatten layers were used. Adam optimizer was used for optimization and the activation function "ReLu" was used. In the final layer, the "Softmax" activation function was used, as it is a multi-class classifier.

The deep learning technique that was used in the making of the model was neural networks. The framework of the model is illustrated in Fig. 7.

2.4 Neural Network Architecture Description

This neural network architecture is a sequential model designed for multi-class classification tasks. It comprises of several fully connected layers with different numbers of units and activation functions. The details of these layers are shown in Fig. 8. The model is compiled with the Adam optimizer, sparse categorical cross-entropy loss function, and accuracy as the evaluation metric.

The input layer of the network takes the shape of the input training data, assuming X_{train} represents the input data. The

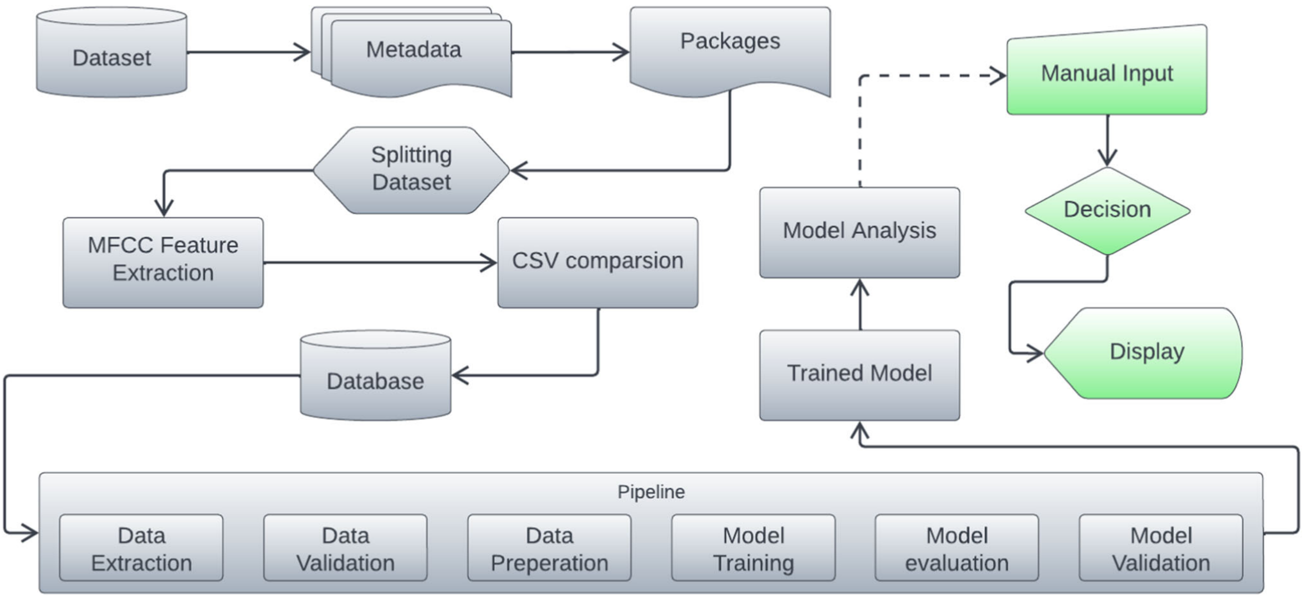

Fig. 7 Implementation and working of the model

network's input layer takes a feature vector of size 40, corresponding to the number of MFCCs extracted from each audio sample. The activation function used in this layer is the rectified linear unit (ReLU), which introduces nonlinearity to the model. Dense layers are used to learn complex patterns and relationships in the data, while dropout layers prevent overfitting by randomly dropping out some units during training, reducing the co-dependence among units and improving generalization. Lastly, a flattened layer is used to reshape the previous layer's output into a one-dimensional vector, which is required for the input of the dense layer.

The model uses the Adam optimizer. Adam is an efficient optimization algorithm that adapts the learning rate during training. The model is optimized based on the sparse categorical cross-entropy loss function, which is suitable for multi-class classification problems. This loss function measures the dissimilarity between the predicted class probabilities and the true class labels. During training, the model's performance is evaluated using the accuracy metric, which measures the percentage of correctly predicted classes. The model aims to maximize this accuracy metric during training. Initially, the dataset and the metadata were imported into the model. Relevant packages were imported for the implementation of the neural network. To verify the accuracy, the dataset was split for training and testing. The MFCC were extracted from both training and testing datasets. The extracted MFCC were compared to those already extracted for the same dataset to label the dataset. The dataset was run through the pipeline of training and testing the model.

```
_________________________________________________________________
Layer (type)                Output Shape              Param #
=================================================================
dense_14 (Dense)            (None, 1024)              14336

dropout_12 (Dropout)        (None, 1024)              0

dense_15 (Dense)            (None, 512)               524800

dropout_13 (Dropout)        (None, 512)               0

dense_16 (Dense)            (None, 256)               131328

dropout_14 (Dropout)        (None, 256)               0

dense_17 (Dense)            (None, 128)               32896

dropout_15 (Dropout)        (None, 128)               0

dense_18 (Dense)            (None, 64)                8256

dropout_16 (Dropout)        (None, 64)                0

dense_19 (Dense)            (None, 32)                2080

dropout_17 (Dropout)        (None, 32)                0

dense_20 (Dense)            (None, 4)                 132

=================================================================
Total params: 713,828
Trainable params: 713,828
Non-trainable params: 0
```

Fig. 8 Neural networks layers

3 Results

From a dataset containing 400 unique audio files, we constructed a model that incorporated a total of 741,476 parameters. Subsequently, we derived the MFCC from these audio files and conducted 200 epochs to enhance the model's accuracy as much as possible. This resulted in a distinct graph of validation error and accuracy. To incorporate multiple

```
[-202.33418     52.436863    -30.788752    -15.815485    -21.129396
  -8.742579     -6.8097987     5.190659    -18.704393     11.076956
 -14.966026     -6.225648    -13.867588     -5.8016       -0.6982322
  20.515596     -8.038788     19.644484      0.32608885    2.0037036
   0.26333922   10.335904     -2.9863083     4.5079513    -4.1324143
  -7.3808284     0.954594     -3.9258196     0.36253187   -7.504233
  -1.7006358     4.0840096     1.8866123    -0.7598902    -3.4504726
  -2.7429857    -1.6647447    -1.9008192     0.38516027    0.75330555]
[[-202.33418     52.436863    -30.788752    -15.815485    -21.129396
   -8.742579     -6.8097987     5.190659    -18.704393     11.076956
  -14.966026     -6.225648    -13.867588     -5.8016       -0.6982322
   20.515596     -8.038788     19.644484      0.32608885    2.0037036
    0.26333922   10.335904     -2.9863083     4.5079513    -4.1324143
   -7.3808284     0.954594     -3.9258196     0.36253187   -7.504233
   -1.7006358     4.0840096     1.8866123    -0.7598902    -3.4504726
   -2.7429857    -1.6647447    -1.9008192     0.38516027    0.75330555]]
(1, 40)
1/1 [==============================] - 0s 22ms/step
[2]
array(['relax'], dtype='<U5')
```

Fig. 9 Prediction of model

reciters, we have used Quranic recitation placed from [21] of six different reciters. For each of these reciter, we extract the acoustic features and classify the audio. We inserted a Quranic verse to be analyzed, the model extracted its MFCC, a part of which is shown in Fig. 10, and predicted its designated emotion and displayed the result as an array illustrated in Fig. 9. The final accuracy of the model turned out to be 77% as shown in Fig. 11. By running multiple Quranic verse through the machine learning code mostly were classified as relaxing.

Based on our findings, we can deduce that using the Mel-frequency cepstral coefficients (MFCC) technique to extract features from audios in the dataset is useful in classifying Quranic verses as calming. The created model extracted 741,476 parameters, allowing for the accurate extraction of MFCCs from the dataset's audios. We were able to obtain a different graph of validation error and accuracy after training the model for 200 epochs, which further confirms the accuracy of our results. The deep learning applications that were used to create the model from the retrieved features are responsible for its correctness.

Furthermore, by introducing a Quranic verse to be analyzed, we were able to extract its MFCC, which supplied us with significant insights into the emotion for which it was allocated. The result was displayed as an array, demonstrating the efficacy of our system for categorizing Quranic verses based on emotional states. Overall, the findings of our study show the potential of employing MFCCs and deep learning algorithms to extract features from audios in the context of Quranic recitation, which could have significant consequences for Quranic studies and the creation of a healing environment.

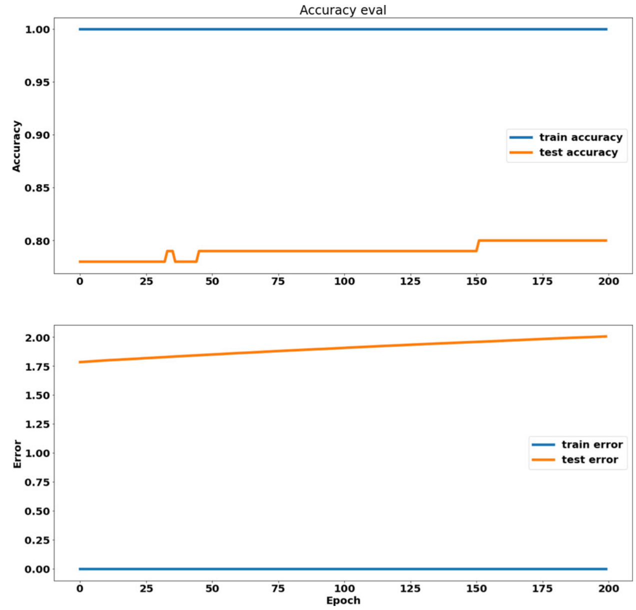

Fig. 11 Accuracy and loss of model

3.1 Comparative Studies

As discussed in Sect. 1, MFCC is widely used for audio feature extraction. Ref. [22] presents two feature extraction methods MFCC and recurrence quantification analysis (RQA). According to the experiment, data extraction using the MFCC method obtains better classification results than the RQA. The prediction value with MFCC reaches 77%. Another work closely related to ours is [23], where the successful detection of stress based on speech signals. Speech signals were captured and analyzed in detecting stress. MFCC feature was extracted from the speech signals. The stress was detected with the neural network that was coded into a program system. Apart from these works, another direction present in literature is to employ electroencephalography (EEG) signal analysis for emotion classification in subjects listening to Quran recitation [24]

mfcc	mfcc2	mfcc3	mfcc4	mfcc5	mfcc6	mfcc7	mfcc8	mfcc9	mfcc10	mfcc11	mfcc12	mfcc13
11.485293	52.694454	6.4672685	25.595886	2.214953	2.8142695	-0.8519743	4.253139	13.443974	15.701662	-17.286722	1.482694	-11.998751
10.666107	50.228428	4.3020315	25.253872	-5.1411967	-1.1816307	-5.6990633	2.6091366	12.691689	19.100025	-15.597666	-0.42082202	-8.505865
-7.280842	45.203884	7.211931	24.059444	-8.941715	-7.5504966	-10.820677	1.8300009	8.01932	14.781019	-17.404804	-11.761877	-14.646868
-23.946793	47.30674	16.30344	24.451649	-3.5835762	-7.5554304	-11.623659	6.6690087	6.450489	8.219208	-17.646976	-18.84282	-16.60274
-14.573497	49.570976	8.38875	24.996624	-8.471899	-4.6146	-16.386839	1.7968928	8.582506	7.958048	-13.234039	-13.160864	-13.30647
2.3833885	52.36577	-5.013781	19.313631	-10.166662	1.3495817	-14.263855	0.7036451	8.215092	11.195814	-8.57453	-5.365735	-19.046186
-6.1780887	57.37553	-4.040032	12.688786	-10.143839	-0.3549705	-12.24184	1.2906312	7.6674657	9.324999	-6.029012	3.7790866	-20.67905
-18.17729	71.87554	1.0481478	-1.5022548	-14.112274	0.4641595	-5.345021	5.263732	2.3507562	3.1608768	-9.169635	8.210405	-15.721555
-16.83406	87.47632	6.189846	-9.627092	-21.307964	-1.0549138	-2.4525142	2.447925	0.4011984	2.6289167	-14.852123	3.048563	-13.819897
-19.948555	87.42551	3.0170984	-5.507799	-21.663097	-4.6988482	-3.2385678	-0.7456046	-2.7290342	4.54632	-9.773302	5.8109345	-10.102326

Fig. 10 MFCC extracted

4 Conclusion

This paper provides empirical data on how audio can be deemed relaxing or soothing. It gives a method to use MFCC for the categorization of different Quranic verses into four distinct emotions based on a set of predetermined audios. The main contribution of this paper is to try and give a scientific explanation to the soothing nature of Quranic verses by outlining the features, which control how an audio is perceived to be of different emotions, by a human being. The goal was to give an unbiased analysis and explain the reasoning behind the emotions induced by Quranic verses to humans by employing MFCC in a machine learning model.

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Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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